1. In active voice, the grammatical subject of the sentence:
- **(A)** Performs the action
- **(B)** Receives the action
- **(C)** Is omitted
- **(D)** Functions as a preposition

2. In passive voice, the grammatical subject of the sentence:
- **(A)** Performs the action
- **(B)** Receives the action
- **(C)** Is always a human agent
- **(D)** Must be plural

3. Identify the sentence written in passive voice:
- **(A)** We pipetted the reagent.
- **(B)** The reagent was pipetted.
- **(C)** The reagent dissolved.
- **(D)** The technician pipetted rapidly.

4. Which auxiliary verb is fundamental to forming all passive voice constructions?
- **(A)** Have
- **(B)** Be
- **(C)** Do
- **(D)** Can

5. What is the passive form of: 'The enzyme cleaves the substrate'?
- **(A)** The substrate is cleaved by enzyme.
- **(B)** The substrate was cleaved by enzyme.
- **(C)** The substrate cleaved the enzyme.
- **(D)** The substrate is cleaving enzyme.

6. What is the passive form of: 'The technician sterilized the glassware'?
- **(A)** Glassware is sterilized.
- **(B)** Glassware was sterilized by technician.
- **(C)** Glassware had sterilized.
- **(D)** Glassware was sterilizing.

7. An agentless (truncated) passive sentence is one where:
- **(A)** The subject is missing
- **(B)** The 'by'-agent phrase is omitted
- **(C)** The verb is active
- **(D)** No object existed

8. Why is the agentless passive dominant in Materials and Methods sections?
- **(A)** To obscure who did the work
- **(B)** To emphasize the objective procedure
- **(C)** Because agents are forbidden
- **(D)** To make sentences longer

9. Which modal auxiliary expresses absolute necessity or strict regulatory command?
- **(A)** May
- **(B)** Might
- **(C)** Must
- **(D)** Could

10. Which modal auxiliary expresses tentative possibility in scientific claims?
- **(A)** Must
- **(B)** May
- **(C)** Shall
- **(D)** Will

11. In 'Bacterial cells can synthesize essential amino acids', 'can' expresses:
- **(A)** Innate physical ability / capacity
- **(B)** Formal regulatory permission
- **(C)** Obligation
- **(D)** Past habitual action

12. Which modal is traditionally used to express formal requirement in patents and standards?
- **(A)** Shall
- **(B)** Might
- **(C)** Could
- **(D)** May

13. Choose the correct passive: 'Scientists are developing a novel biosensor.'
- **(A)** A novel biosensor is developed.
- **(B)** A novel biosensor is being developed.
- **(C)** A novel biosensor was developed.
- **(D)** A novel biosensor has developed.

14. Choose the correct passive: 'The team has synthesized three peptide analogs.'
- **(A)** Three peptide analogs have synthesized.
- **(B)** Three analogs have been synthesized.
- **(C)** Three analogs were synthesized.
- **(D)** Three analogs had synthesized.

15. Choose the correct modal for logical deduction: 'The culture is turbid; bacteria ___ be growing.'

(A) must
(B) might not
(C) shall
(D) ought not

16. Which modal indicates advice or standard operational recommendation?

(A) Should
(B) Shall
(C) Must
(D) May

17. In 'You may now remove the tubes from the thermal cycler', 'may' expresses:

(A) Permission
(B) Ability
(C) Obligation
(D) Past condition

18. What is the passive form of: 'Researchers will evaluate the pharmacokinetic profile'?

(A) The profile will evaluate.
(B) The profile will be evaluated.
(C) The profile will have evaluated.
(D) The profile is evaluated.

19. Which sentence is an impersonal passive construction?

(A) It was observed that cells lysed.
(B) We observed that cells lysed.
(C) Cells lysed rapidly.
(D) Observations were made by us.

20. Which modal expresses past unfulfilled possibility?

(A) Could have
(B) Must
(C) Shall
(D) Will

21. Choose the correct modal: 'The sample ___ have been contaminated during vortexing.'

(A) might
(B) shall
(C) will
(D) can

22. Can intransitive verbs (e.g., deteriorate, arrive, sleep) be converted into passive voice?

(A) Yes, always
(B) No, never
(C) Only in present tense
(D) Only in past tense

23. Why cannot intransitive verbs be converted into passive voice?

(A) They lack a direct object to become subject
(B) They lack auxiliary verbs
(C) They are always stative
(D) They are irregular

24. What is the passive form of: 'The centrifuge had spun the tubes for 10 minutes'?

(A) The tubes had been spun.
(B) The tubes were spun.
(C) The tubes have been spun.
(D) The tubes had spun.

25. Which modal auxiliary denotes past habit or routine action in research?

(A) Would
(B) Must
(C) Shall
(D) Can

26. In 'The mutation could disrupt the active site', 'could' functions as an:

(A) Epistemic hedge of possibility
(B) Absolute statement of fact
(C) Imperative command
(D) Historical narrative marker

27. Which semi-modal is synonymous with 'must' in expressing external obligation?

(A) Have to
(B) Ought to
(C) Used to
(D) Dare to

28. Choose the correct negative modal for prohibition in biosafety:

(A) You do not have to enter.
(B) You must not enter without PPE.
(C) You might not enter.
(D) You could not enter.

29. In 'The results should be reproducible under standard conditions', 'should' expresses:
- **(A)** Reasonable expectation
- **(B)** Physical ability
- **(C)** Strict legal obligation
- **(D)** Past regret

30. What is the passive form of modal verbs: 'The team must autoclave all biohazardous waste'?
- **(A)** Waste must autoclave.
- **(B)** Waste must be autoclaved.
- **(C)** Waste must have autoclaved.
- **(D)** Waste was autoclaved.

31. Identify the modal of deduction indicating high improbability:
- **(A)** Cannot
- **(B)** Must
- **(C)** May
- **(D)** Should

32. In 'This band cannot be the target amplicon because its size is 1,000 bp', 'cannot' expresses:
- **(A)** Logical impossibility
- **(B)** Lack of physical skill
- **(C)** Prohibition
- **(D)** Permission

33. What is the passive form of: 'The enzyme can hydrolyze cellulose'?
- **(A)** Cellulose can be hydrolyzed.
- **(B)** Cellulose can hydrolyze.
- **(C)** Cellulose could be hydrolyzed.
- **(D)** Cellulose was hydrolyzed.

34. Which of the following sentences uses active voice effectively to highlight agency?
- **(A)** We designed the primers to anneal at 60 deg C.
- **(B)** The primers were designed to anneal.
- **(C)** Designing was performed on primers.
- **(D)** Primer design took place.

35. In 'Cells ought to be incubated immediately', 'ought to' conveys:
- **(A)** Moral/operational advisability
- **(B)** Innate physical power
- **(C)** Pure speculative doubt
- **(D)** Future prediction

36. Choose the correct passive: 'Someone left the incubator door open.'
- **(A)** The incubator door was left open.
- **(B)** The incubator door is left open.
- **(C)** The incubator door has left open.
- **(D)** The incubator door was being left.

37. Which modal replaces 'can' in past tense to denote general past ability?
- **(A)** Could
- **(B)** Might
- **(C)** Should
- **(D)** Would

38. Choose the correct phrase: 'The researchers were ___ isolate the intact plasmid.'
- **(A)** able to
- **(B)** can
- **(C)** could to
- **(D)** must to

39. In 'The protocol might require optimization', 'might' indicates:
- **(A)** Tentative speculative possibility
- **(B)** Guaranteed outcome
- **(C)** Legal necessity
- **(D)** Complete impossibility

40. Identify the sentence that contains both a modal auxiliary and passive voice:
- **(A)** The reaction should be terminated after 15 minutes.
- **(B)** The reaction terminated after 15 minutes.
- **(C)** We should terminate the reaction.
- **(D)** The reaction has terminated.

41. Select the sentence where passive voice is rhetorically preferred over active voice:
- **(A)** A technician washed the beaker.
- **(B)** The beaker was washed with chromic acid.
- **(C)** We added 5 mL of acid.
- **(D)** Someone calibrated the meter.

42. Which sentence demonstrates proper scientific hedging using modal verbs?
- **(A)** This mutation causes cancer in humans.
- **(B)** This mutation may contribute to carcinogenesis.
- **(C)** This mutation definitely causes cancer.
- **(D)** This mutation will prove to cause cancer.

43. What is the passive voice of: 'The Principal Investigator gave the postdoc a new plasmid'?
- **(A)** A new plasmid was given to the postdoc.
- **(B)** The postdoc was given a new plasmid.
- **(C)** Both A and B are grammatically valid.
- **(D)** Neither A nor B is correct.

44. Why can ditransitive verbs (give, assign, grant) generate two passive forms?
- **(A)** They possess both a direct and an indirect object
- **(B)** They are irregular verbs
- **(C)** They contain two finite predicates
- **(D)** They take modal auxiliaries

45. Choose the correct modal: 'The frozen cells ___ thawed rapidly at 37 deg C to preserve viability.'
- **(A)** must be
- **(B)** might be
- **(C)** can be
- **(D)** could be

46. Identify the sentence using an impersonal passive reporting structure:
- **(A)** It is believed that mitochondria originated via endosymbiosis.
- **(B)** Biologists believe mitochondria originated.
- **(C)** Mitochondria originated via endosymbiosis.
- **(D)** Endosymbiosis is believed.

47. Choose the correct passive: 'Nobody has ever observed this anomalous phenotype.'
- **(A)** This anomalous phenotype has never been observed.
- **(B)** This phenotype was never observed.
- **(C)** This phenotype had never observed.
- **(D)** This phenotype is never observed.

48. In 'The protein would precipitate if salt were added', 'would' is used as:
- **(A)** A conditional consequence modal
- **(B)** A past habit marker
- **(C)** A polite request
- **(D)** An imperative

49. Choose the correct past modal: 'We didn't detect the band; the probe ___ degraded.'
- **(A)** must have
- **(B)** should have
- **(C)** can have
- **(D)** shall have

50. In 'The probe must have degraded', 'must have' indicates:
- **(A)** A confident logical deduction about a past event
- **(B)** A past obligation that was neglected
- **(C)** A past ability
- **(D)** A future projection

51. Choose the correct past modal: 'You ___ checked the pH before adding the enzyme; it was too acidic!'
- **(A)** should have
- **(B)** must have
- **(C)** might have
- **(D)** would have

52. In 'You should have checked the pH', 'should have' expresses:
- **(A)** Criticism of a neglected past duty
- **(B)** A logical certainty
- **(C)** A past physical ability
- **(D)** A successful completed action

53. Select the sentence with an ergative verb used intransitively without passive marking:
- **(A)** The cell membrane ruptured under osmotic stress.
- **(B)** The cell membrane was ruptured by osmotic stress.
- **(C)** Osmotic stress ruptured the cell membrane.
- **(D)** Both A and C are grammatically active.

54. Choose the correct passive: 'They are to publish the clinical trial protocol next month.'
- **(A)** The protocol is to be published next month.
- **(B)** The protocol was published next month.
- **(C)** The protocol will be publish next month.
- **(D)** The protocol has to be publish.

55. Identify the sentence that improperly uses passive voice, producing unnecessary wordiness:
- **(A)** A determination of the kinetic parameters was performed by us.
- **(B)** We determined the kinetic parameters.
- **(C)** Kinetic parameters were determined.
- **(D)** The kinetic parameters were evaluated.

56. Choose the correct modal: 'All biosafety cabinets ___ undergo annual HEPA recertification.'
- **(A)** must
- **(B)** might
- **(C)** could
- **(D)** may

57. In 'Preliminary data indicate that compound X could potentially inhibit viral fusion', what is stylistically flawed?

(A) 'Could potentially' is an ungrammatical double hedge
(B) Compound X must be capitalized
(C) Indicate requires past tense
(D) Inhibit is intransitive

58. Select the sentence where active voice is preferred to emphasize authorial responsibility:

(A) In this paper, we argue that current viral vectors are inefficient.
(B) In this paper, it is argued that current vectors are inefficient.
(C) Current vectors are argued to be inefficient.
(D) Arguments are made by us regarding vectors.

59. What is the passive form of: 'The lab must have discarded the expired reagents'?

(A) The expired reagents must have been discarded.
(B) The expired reagents must be discarded.
(C) The expired reagents had been discarded.
(D) The expired reagents were discarded.

60. Choose the correct modal: 'You ___ not touch the acrylamide gel without nitrile gloves!'

(A) must
(B) might
(C) could
(D) would

61. Identify the sentence where 'can' expresses epistemic possibility rather than dynamic ability:

(A) Contamination can ruin months of cell culture work.
(B) The technician can pipette 0.1 uL accurately.
(C) The autoclave can hold twelve flasks.
(D) She can speak scientific French.

62. Choose the correct passive: 'Researchers have to monitor bioreactor temperature constantly.'

(A) Temperature has to be monitored constantly.
(B) Temperature had to be monitored.
(C) Temperature is to monitor constantly.
(D) Temperature has been monitored.

63. In 'The enzyme is thought to interact with membrane lipids', what passive construction is used?

(A) Personal passive with infinitive complement
(B) Impersonal passive with dummy 'it'
(C) Truncated imperative
(D) Active intransitive

64. Transform into active voice: 'A significant increase in absorbance was observed by the team.'

(A) The team observed a significant increase in absorbance.
(B) Absorbance increased significantly.
(C) Observing absorbance, the team worked.
(D) The team was observing absorbance.

65. Choose the correct modal: 'If the primer is degraded, the PCR ___ fail.'

(A) may
(B) must
(C) shall
(D) can

66. Select the sentence with a pseudo-passive (prepositional passive):

(A) The experimental protocol was thoroughly agreed upon.
(B) The experimental protocol was agreed.
(C) The protocol was written.
(D) The protocol was published.

67. Why is 'The protocol was agreed upon' grammatically valid in English?

(A) English permits preposition stranding in prepositional passives
(B) Agreed is transitive
(C) Upon is an adverb
(D) It is informal slang

68. Choose the correct modal: 'Cryo-EM ___ resolve macromolecular structures at near-atomic resolution.'

(A) can
(B) must
(C) shall
(D) ought

69. Identify the sentence that correctly uses passive voice with a causative verb:

(A) The cells were made to express the recombinant antigen.
(B) The cells were made express the recombinant antigen.
(C) The cells made to express the antigen.
(D) The cells were let express the antigen.

70. In active voice, 'make' takes a bare infinitive ('We made the cells express'); in passive, it takes:

(A) A full 'to'-infinitive ('The cells were made to express')
(B) A bare infinitive
(C) A gerund
(D) A past participle

71. Choose the correct modal: 'The autoclave steam valve ___ be inspected weekly per OSHA standards.'

(A) shall
(B) might
(C) could
(D) may

72. Identify the sentence containing an erroneous double modal:

(A) The team might could finish the assay today.
(B) The team might be able to finish the assay today.
(C) The team may finish the assay today.
(D) The team could finish the assay today.

73. Why is 'might could' ungrammatical in standard English syntax?

(A) Standard English prohibits stacking two modal auxiliary verbs in one verb phrase
(B) Could is present tense
(C) Might is plural
(D) Finish requires a gerund

74. Choose the correct passive: 'One can deduce the molecular weight from migration distance.'

(A) The molecular weight can be deduced from migration distance.
(B) Molecular weight could deduce.
(C) Molecular weight is deduced by one.
(D) Migration distance can deduce molecular weight.

75. Select the sentence that avoids misplaced agency in passive construction:

(A) The cultures were incubated at 37 deg C by the thermostat.
(B) The cultures were incubated at 37 deg C.
(C) The thermostat incubated the cultures.
(D) Incubation was performed by the thermostat.

76. Identify the sentence where passive voice introduces dangerous syntactic ambiguity:

(A) The rats were injected with the virus, and they were examined after 24 hours.
(B) The virus was injected into the rats, and they were examined after 24 hours.
(C) After virus injection, examination of rats occurred.
(D) Having injected the virus, rats were examined.

77. Select the sentence that deploys epistemic modality with maximal academic sophistication:

(A) These empirical findings strongly suggest that kinase activation may be contingent upon scaffold dimerization.
(B) These empirical findings prove that kinase activation must be contingent upon dimerization.
(C) These empirical findings will definitely show that kinase is activated.
(D) These findings show that kinase activation is probably contingent.

78. Choose the correct modal for a hypothetical retrospective concession: 'Even if the probe had bound, the signal ___ have been undetectable due to quenching.'

(A) might
(B) shall
(C) must
(D) can

79. What is the passive form of: 'The Principal Investigator made the graduate student re-run the electrophoresis gel'?

(A) The graduate student was made to re-run the electrophoresis gel.
(B) The graduate student was made re-run the electrophoresis gel.
(C) The gel was made to re-run by the student.
(D) The graduate student made the gel re-run.

80. Select the sentence where active voice is essential to prevent a dangling agent:

(A) To isolate the mitochondria, the tissue was homogenized.
(B) To isolate the mitochondria, we homogenized the tissue.
(C) For isolating mitochondria, homogenization was carried out.
(D) Homogenization of tissue was performed to isolate mitochondria.

81. In 'It is recommended that the reaction mixture ___ incubated on ice':

(A) be
(B) is
(C) was
(D) are

82. Choose the correct modal expression: 'The laboratory ___ decontaminate all surfaces before exiting BSL-3.'

(A) is required to
(B) can to
(C) may to
(D) might to

83. Identify the sentence containing an erroneous passive transformation of a stative verb:

(A) The flask was contained 500 mL of broth.
(B) The flask contained 500 mL of broth.
(C) 500 mL of broth was contained in the flask.
(D) Broth was held in the flask.

84. Why cannot 'The flask was contained 500 mL' be accepted?

(A) 'Contain' is an active stative verb; inserting 'was contained' produces ungrammatical passive syntax
(B) Flask is inanimate
(C) Broth is uncountable
(D) 500 mL is plural

85. Choose the correct past modal: 'The wild-type colonies grew, so the ampicillin plate ___ lacked the antibiotic.'

(A) must have
(B) should have
(C) can have
(D) shall have

86. Select the sentence that effectively mitigates over-claiming (hedging) in Discussion sections:

(A) Our data demonstrate that protein X is the sole regulator of apoptosis.
(B) Our data suggest that protein X is an important regulator of apoptosis.
(C) Our data prove beyond doubt that protein X regulates apoptosis.
(D) Protein X is the only regulator of apoptosis as proven by our data.

87. What is the passive form of: 'They let the culture incubate for 48 hours'?

(A) The culture was allowed to incubate for 48 hours.
(B) The culture was let incubate for 48 hours.
(C) The culture was let to incubate for 48 hours.
(D) The culture let incubated for 48 hours.

88. Why is 'The culture was let incubate' avoided in formal English?

(A) The verb 'let' resists passivization; standard syntax replaces it with 'be allowed to'
(B) 'Let' is always past tense
(C) 'Incubate' is transitive
(D) 48 hours is plural

89. Choose the correct modal: 'The bio-sensor ___ detect antigens at concentrations as low as 1 pg/mL.'

(A) is able to
(B) shall to
(C) must to
(D) may to

90. Select the sentence with a dangling passive infinitive:

(A) To assess cytotoxicity, the MTT assay was performed.
(B) To assess cytotoxicity, we performed the MTT assay.
(C) To assess cytotoxicity, the researchers performed the MTT assay.
(D) In assessing cytotoxicity, the MTT assay was used by our team.

91. Choose the correct modal: 'Judging from the sharp melting peak, the amplicon ___ be highly specific.'

(A) must
(B) might not
(C) shall
(D) can

92. Identify the sentence that correctly transforms an indirect object into a passive subject:

(A) The postdoctoral scholar was awarded a research fellowship.
(B) A research fellowship was awarded to the postdoctoral scholar.
(C) Both A and B are grammatically impeccable.
(D) Neither A nor B is standard.

93. Choose the correct modal: 'The researcher ___ have calibrated the balance; all weighings are flawed!'

(A) ought to
(B) must
(C) can
(D) shall

94. In 'The protein was found to be phosphorylated at Ser-128', what is the active equivalent?

(A) Researchers found that the protein was phosphorylated at Ser-128.
(B) The protein found phosphorylation at Ser-128.
(C) Phosphorylation found the protein at Ser-128.
(D) It was found the protein phosphorylated.

95. Select the sentence that avoids an ungrammatical 'by'-phrase with an inanimate instrument:

(A) The membrane was severed by a scalpel.
(B) The membrane was severed with a scalpel.
(C) The membrane was severed through a scalpel.
(D) The membrane was severed of a scalpel.

96. Why is 'with a scalpel' grammatically correct while 'by a scalpel' is flawed?

(A) 'By' denotes the causative agent, while 'with' denotes the physical instrument
(B) Scalpel is singular
(C) Severed is irregular
(D) Membrane is biological

97. Choose the correct modal: 'Had the enzyme been folded, it ___ have exhibited catalytic turnover.'

(A) would
(B) will
(C) shall
(D) can

98. Identify the sentence demonstrating flawless integration of modal hedging and passive voice:

(A) It may be hypothesized that off-target cleavage was mitigated by modified guide chemistry.
(B) It will be hypothesized that off-target cleavage was mitigated.
(C) We must hypothesize that off-target cleavage mitigated.
(D) Off-target cleavage must have mitigated by guide chemistry.

99. Choose the correct modal: 'The bio-safety cabinet airflow ___ not be obstructed under any operational circumstances.'

(A) must
(B) might
(C) could
(D) would

100. Identify the sentence that achieves impeccable syntactic balance between active authorial agency and passive methodological objectivity:

(A) Although we initially hypothesized that the promoter was constitutively active, subsequent reporter assays confirmed that transcription is strictly repressed in the absence of exogenous ligand.
(B) Although the promoter was initially hypothesized to be constitutively active by us, it was confirmed by reporter assays that transcription is repressed.
(C) We hypothesized the promoter and we ran reporter assays and we found repression.
(D) Hypothesizing promoter constitutive activity, reporter assays were conducted.

SECTION B: DESCRIPTIVE & SUBJECTIVE QUESTIONS

Q1. Analyze the syntactic mechanics of voice transformation from active to passive across the full spectrum of English tenses.

Q2. Examine the rhetorical, scientific, and cognitive justifications for the dominance of the agentless passive in experimental methodologies.

Q3. Discuss the legitimate role of the active voice in contemporary scientific papers, analyzing guidelines from Nature, Science, and NIH.

Q4. Analyze impersonal passive constructions ('It is reported that...', 'It was observed that...') and evaluate their stylistic impact.

Q5. Examine the taxonomy of English modal auxiliaries, distinguishing epistemic, deontic, and dynamic modalities in science.

Q6. Analyze modal hedging techniques in scientific Results and Discussion sections to calibrate empirical claims.

Q7. Detail the syntax and semantics of past modal constructions (Could have, Might have, Must have, Should have) in experimental troubleshooting.

Q8. Examine the syntax of ditransitive verbs in passive voice, analyzing direct versus indirect object passivization.

Q9. Analyze prepositional passives (pseudo-passives) and the phenomenon of preposition stranding in technical prose.

Q10. Examine ergative (middle) verbs versus passive voice in describing biochemical and biophysical state changes.

Q11. Discuss the Mandative Subjunctive versus modal periphrastics ('should') in international biosafety standards.

Q12. Analyze the misuse of 'can' versus 'may' in scientific laboratory permissions and protocol documentation.

Q13. Examine passive transformations of causative verbs ('make', 'let', 'have') in experimental methodology descriptions.

Q14. Analyze the syntax and semantics of 'Must not' versus 'Do not have to' in biosafety containment levels.

Q15. Examine the dangling agent problem in passive constructions with introductory purpose or participial phrases.

Q16. Analyze the epistemic strength of modal verbs in constructing formal scientific hypotheses.

Q17. Detail the passive voice of phrasal verbs, prepositional verbs, and multi-word verbs in biotechnology literature.

Q18. Examine the passive voice with modal perfects (Must have been, Could have been, Should have been) in retrospective audit reports.

Q19. Discuss the stylistic peril of 'Passive Stacks' (chained passive clauses) and demonstrate sentence disentangling.

Q20. Analyze modal verbs of deduction in interpreting negative controls and experimental anomalies.

Q21. Examine the syntax of 'Need' and 'Dare' as marginal modal auxiliaries versus lexical main verbs.

Q22. Analyze the passive voice in describing flowcharts, manufacturing pipelines, and bioprocess engineering flow.

Q23. Examine the modal 'Would' in describing past laboratory routines, prospective designs, and polite discourse.

Q24. Analyze the passive transformation of verbs taking that-clause complements and infinitive raising.

Q25. Examine the modal auxiliary 'Shall' in legal patents, regulatory compliance, and scientific declarations.

Q26. Analyze the interaction between negation and modal verbs (Epistemic vs. Deontic Scope of Negation).

Q27. Discuss the passive voice in writing Material Transfer Agreements (MTAs) and Non-Disclosure Agreements (NDAs).

Q28. Examine the avoidance of passive voice with reciprocal verbs and non-passivizable idioms in technical discourse.

Q29. Analyze the modal verb 'Used to' versus 'Would' in historical accounts of biotechnological breakthroughs.

Q30. Synthesize guidelines for achieving stylistic harmony between active voice, passive voice, and modal hedging in a doctoral dissertation.

SECTION C: ANSWER KEY - MULTIPLE CHOICE QUESTIONS

Q1:	[A]	**Q21:**	[A]	**Q41:**	[B]	**Q61:**	[A]	**Q81:**	[A]
Q2:	[B]	**Q22:**	[B]	**Q42:**	[B]	**Q62:**	[A]	**Q82:**	[A]
Q3:	[B]	**Q23:**	[A]	**Q43:**	[C]	**Q63:**	[A]	**Q83:**	[A]
Q4:	[B]	**Q24:**	[A]	**Q44:**	[A]	**Q64:**	[A]	**Q84:**	[A]
Q5:	[A]	**Q25:**	[A]	**Q45:**	[A]	**Q65:**	[A]	**Q85:**	[A]
Q6:	[B]	**Q26:**	[A]	**Q46:**	[A]	**Q66:**	[A]	**Q86:**	[B]
Q7:	[B]	**Q27:**	[A]	**Q47:**	[A]	**Q67:**	[A]	**Q87:**	[A]
Q8:	[B]	**Q28:**	[B]	**Q48:**	[A]	**Q68:**	[A]	**Q88:**	[A]
Q9:	[C]	**Q29:**	[A]	**Q49:**	[A]	**Q69:**	[A]	**Q89:**	[A]
Q10:	[B]	**Q30:**	[B]	**Q50:**	[A]	**Q70:**	[A]	**Q90:**	[A]
Q11:	[A]	**Q31:**	[A]	**Q51:**	[A]	**Q71:**	[A]	**Q91:**	[A]
Q12:	[A]	**Q32:**	[A]	**Q52:**	[A]	**Q72:**	[A]	**Q92:**	[C]
Q13:	[B]	**Q33:**	[A]	**Q53:**	[D]	**Q73:**	[A]	**Q93:**	[A]
Q14:	[B]	**Q34:**	[A]	**Q54:**	[A]	**Q74:**	[A]	**Q94:**	[A]
Q15:	[A]	**Q35:**	[A]	**Q55:**	[A]	**Q75:**	[B]	**Q95:**	[B]
Q16:	[A]	**Q36:**	[A]	**Q56:**	[A]	**Q76:**	[A]	**Q96:**	[A]
Q17:	[A]	**Q37:**	[A]	**Q57:**	[A]	**Q77:**	[A]	**Q97:**	[A]
Q18:	[B]	**Q38:**	[A]	**Q58:**	[A]	**Q78:**	[A]	**Q98:**	[A]
Q19:	[A]	**Q39:**	[A]	**Q59:**	[A]	**Q79:**	[A]	**Q99:**	[A]
Q20:	[A]	**Q40:**	[A]	**Q60:**	[A]	**Q80:**	[B]	**Q100:**	[A]

MCQ Scoring Guide & Difficulty Distribution:

Questions 1 to 40: Foundational & Conceptual [Easy] | Questions 41 to 75: Analytical & Applied [Moderate]

Questions 76 to 100: Complex Numerical & Advanced Problem Solving [Difficult]

- Preserving Authorial Anonymity in Blind Peer Review:
- Agentless passives prevent accidental revelation of laboratory identities in double-blind submission processes.

>>Solution to Question 8: Use the passive voice to emphasize the action rather than the doer. It is commonly observed that... and evaluated by the National Science and NIH.

- Overuse of impersonal passives clutters text with empty introductory phrases:
- Bloated: "It was observed by the investigators that the cells underwent apoptosis."
- Streamlined Active: "The cells underwent apoptosis."
- Streamlined Methodological: "Apoptosis was observed in 80% of treated cells."

>>Solution Question 6: Analyze the taxonomy of Emotive modal auxiliaries. Discuss instances of epistemic modality in science.

Combining multiple hedging devices in the same clause produces weak, unscholarly prose:

- Flawed: "These data might possibly perhaps indicate a potential binding interaction."
- Revised: "These data suggest that a binding interaction may occur."

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thematic topic of the preceding discourse.

>>**Solution** **Question 20** Middle verbs versus passives (active in construction but passive in meaning) are the construction of verbs with a reflexive pronoun that is structurally active but semantically passive.

In active voice, 'make', 'let', and 'have' govern an object followed by a BARE INFINITIVE (without 'to'):

- "We made the cells express GFP."
- "The technician let the agar solidify."
- "The PI had the postdoc sequence the clones."

2. Morphosyntactic Realignment in Passive Voice:

- Passive of 'Make' (Requires restoring 'to'-infinitive):
- Active: "The osmotic shock made the bacterial cells burst."
- Passive: "The bacterial cells were made to burst by the osmotic shock." (Mandatory 'to').
- Passive of 'Let' (Suppletive replacement with 'allow to'):
- 'Let' cannot be passivized directly (*The agar was let solidify is ungrammatical).
- Passive: "The agar was allowed to solidify for 30 minutes."
- Passive of 'Have' (Suppletive replacement with 'instruct to' or 'ask to'):
- Passive: "The postdoc was instructed to sequence the clones."

>>Solution Step 1: Analyze the syntax and semantics of the passive construction to identify any constraints or limitations on its use in certain contexts or with certain verbs or phrases.

3. Three Remediation Strategies:
- Strategy 1 (Active Voice with Human Agent):
- "To assess microbial viability, the researchers incubated the agar plates for 24 hours."
- Strategy 2 (Recasting Introductory Phrase with Explicit Passive):
- "For microbial viability to be assessed, the agar plates were incubated for 24 hours."
- Strategy 3 (Nominalization of Purpose):
- "Assessment of microbial viability was achieved by incubating the agar plates for 24 hours."

>>Solution Question 14: The passive voice and modal verbs in technical texts, particularly in biotechnology literature,

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2. Modal Formulations in Control Interpretation:
- Deduction of Contamination (Positive Signal in Negative Control):
- "Because the non-template control lane produced a bright 500 bp band, the PCR master mix must be contaminated with exogenous plasmid template." (High certainty; 'must be').
- Deduction of Reagent Failure (Negative Signal in Positive Control):
- "Because the positive control failed to fluoresce, the primary antibody must have lost its binding activity."
- Deduction of Impossibility (Cannot / Can't):
- "The observed 25 kDa band cannot be the full-length recombinant fusion protein, which has a calculated molecular weight of 72 kDa." (Logical impossibility).
- Tentative Inference (May / Might):
- "The faint band at 25 kDa may represent a proteolytic degradation fragment or an alternative splice variant."

>>Solution to Question 22: Here is a flowchart illustrating the passive voice and modal auxiliaries used in bioprocess engineering flow.

vessel."
- "Glucose feed and dissolved oxygen levels are maintained via automated feedback loops."
- "Following stationary phase arrest, the broth is clarified by continuous centrifugation."
- "Target antibodies are subsequently captured using Protein A affinity chromatography, after which viral inactivation is performed at low pH."

>>Solution Question 25: Rewrite the modal "would" in the following sentence to express a polite request: "Would you please pass the salt?" **Solution:** The modal "would" is used to express a polite request. The sentence can be rewritten as: "Could you please pass the salt?" or "Can you please pass the salt?" Both alternatives maintain the polite tone of the original sentence.

In contemporary spoken and academic English, 'shall' has virtually disappeared as a first-person future marker, replaced universally by 'will'.

2. The Specialized Bastion of 'Shall': Legal Contracts & Patent Claims:
In regulatory frameworks (FDA, EMA), international treaties, and patent claims, 'shall' carries immense, legally binding deontic force:
- Mandatory Patent Requirement: "The lyophilized formulation shall contain between 5% and 10% trehalose by weight."
- Legal Meaning: A formulation with 4.9% trehalose falls outside patent coverage; 'shall' establishes strict legal boundary conditions.
- Statutory Regulatory Compliance: "The sponsor shall submit annual safety reports to the regulatory authority."
- Meaning: Absolute legal obligation; failure constitutes a statutory offense.

>>Solution to Question 35: **Solution to Question 36:** Material Transfer Agreements (MTAs) and Non-Disclosure Agreements (NDAs).

- "Upon termination of the research project, all remaining aliquots of the Material shall be autoclaved and destroyed, or returned to the Provider upon request."
- Confidentiality and Intellectual Property:
- "Confidential Information shall be maintained under strict secrecy, and shall be disclosed only to employees with a verified need to know."
- Function: Establishes impersonal, institutional obligations binding upon corporations rather than individual transient researchers.

>>Solution to Question 20: Analyze the modal verb "shall" in the following sentence from a boilerplate clause: "The Receiving Party shall not disclose any Confidential Information to third parties." Explain why "shall" is used instead of "will" or "must" in this context, and discuss its historical origins in legal drafting.

[Problem Statement]: Synthesize guidelines for achieving stylistic harmony between active voice, passive voice, and modal hedging in a doctoral
[Detailed Step-by-Step Answer]:
1. The Rhetorical Symphony of a Dissertation:
A doctoral thesis in biotechnology is an advanced scientific argument requiring a calibrated blend of active agency, objective passivity, and epistemic modesty:

2. Section-by-Section Best Practices:
- Abstract & Title: Active and direct. State the core finding immediately: "Here, we show that..."
- Introduction: Active voice when positioning your doctoral hypothesis against literature; passive when summarizing broad established scientific paradigms.
- Materials and Methods: 85-90% agentless passive voice to ensure total methodological objectivity and universal reproducibility; active voice only when detailing novel customized protocols.
- Results: Balanced alternation. Passive for procedural setups ("Cells were treated with compound 4b"); Active for observational outcomes ("The compound elicited a 3-fold reduction in viability; Figure 3 illustrates this trend").
- Discussion & Conclusions: Modals and hedging become paramount (may, could, should, suggest, indicate) to interpret biological implications with scholarly rigor, avoiding dogmatic over-claiming while actively defending authorial interpretations.
